EC 441 | NETWORK LAYER

# Subnetting & CIDR

Problem Set with Worked Solutions


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## CIDR Notation & Subnet Masks

An IP address in CIDR notation is written as `a.b.c.d/n`, where `/n` indicates the number of prefix (network) bits. The remaining $32 - n$ bits are host bits.

| | |
|---|---|
| **Addresses in block** | $2^{32-n}$ |
| **Usable hosts** | $2^{32-n} - 2$ (minus network & broadcast) |
| **Subnet mask** | $n$ ones followed by $(32 - n)$ zeros |
| **Network address** | IP AND mask (all host bits = 0) |
| **Broadcast address** | Network addr with all host bits = 1 |
| **First / Last usable** | Network +1 / Broadcast −1 |

## Problem 1 | Subnet Basics

Given the address `192.168.10.75/26`, determine:

(a) The subnet mask in dotted decimal
(b) The network (subnet) address
(c) The broadcast address
(d) The range of usable host addresses
(e) The total number of usable hosts

## Solution

**(a) Subnet mask:** /26 means 26 ones, then 6 zeros:

$$\underbrace{11111111.11111111.11111111.11}_{26}\underbrace{000000}_{6} = \texttt{255.255.255.192}$$

**(b) Network address:** AND the IP with the mask. Focus on the last octet:

$$75 = 01001011_2, \quad \text{mask} = 11000000_2$$

$$01001011 \text{ AND } 11000000 = 01000000_2 = 64 \quad \Rightarrow \quad \boxed{\texttt{192.168.10.64}}$$

**(c) Broadcast address:** Set all 6 host bits to 1:

$$01000000 \text{ OR } 00111111 = 01111111_2 = 127 \quad \Rightarrow \quad \boxed{\texttt{192.168.10.127}}$$

**(d) Usable range:** Network $+1$ to Broadcast $-1$:

$$\boxed{\texttt{192.168.10.65}} \text{ to } \boxed{\texttt{192.168.10.126}}$$

**(e) Usable hosts:** $2^6 - 2 =$ `62 hosts`.

## Problem 2 | Designing Subnets

A company is assigned the address block `10.50.0.0/16`. They need subnets for:

| Department | Required Hosts |
|---|---|
| Engineering | 500 |
| Marketing | 120 |
| Sales | 60 |
| Management | 25 |

(a) What prefix length is needed for each department?
(b) Assign non-overlapping subnets, starting from `10.50.0.0`, largest first.
(c) How many addresses are wasted in each subnet?

## Solution

**(a) Prefix lengths:** Find the smallest $2^k \geq (\text{hosts} + 2)$:

| Dept | Need | Block Size | Prefix |
|---|---|---|---|
| Engineering | 500 | $2^{10} = 1024$ | /22 |
| Marketing | 120 | $2^7 = 128$ | /25 |
| Sales | 60 | $2^7 = 128$ | /25 |
| Management | 25 | $2^5 = 32$ | /27 |

> **Why /22 and not /23?** Engineering needs $500 + 2 = 502$ addresses. $2^9 = 512$ seems enough, but 512 total − 2 reserved $= 510$ usable. That works! Actually, /23 gives $2^9 = 512$ total, so 510 usable hosts $> 500$. However, the standard approach uses $2^{10}$ to be safe. *If your professor accepts /23 with the reasoning that $510 \geq 500$, that's also correct.*

**(b) Subnet allocation** (each block must start at a multiple of its size):

| | | |
|---|---|---|
| **Engineering /22** | `10.50.0.0/22` | `10.50.0.0 -- 10.50.3.255` |
| **Marketing /25** | `10.50.4.0/25` | `10.50.4.0 -- 10.50.4.127` |
| **Sales /25** | `10.50.4.128/25` | `10.50.4.128 -- 10.50.4.255` |
| **Management /27** | `10.50.5.0/27` | `10.50.5.0 -- 10.50.5.31` |

**(c) Wasted addresses** = (usable) − needed:

| Dept | Usable | Needed | Wasted |
|---|---|---|---|
| Engineering | 1022 | 500 | 522 |
| Marketing | 126 | 120 | 6 |
| Sales | 126 | 60 | 66 |
| Management | 30 | 25 | 5 |

Total waste: 599 addresses. Engineering dominates due to the jump from $2^9 \rightarrow 2^{10}$.

## Problem 3 | Longest Prefix Match

A router has the following forwarding table:

| Prefix | Next Hop |
|---|---|
| `172.16.0.0/12` | Interface A |
| `172.16.0.0/16` | Interface B |
| `172.16.4.0/22` | Interface C |
| `0.0.0.0/0` | Interface D (default) |

For each destination, determine which interface the packet is forwarded to:

(a) `172.16.4.15` (b) `172.16.5.200` (c) `172.16.128.1` (d) `172.17.0.1` (e) `10.0.0.1`

## Solution

For each address, check all matching prefixes and pick the **longest** (most specific) one.

**(a) 172.16.4.15:**

/12: 172.16.0.0–172.31.255.255 ✓
/16: 172.16.0.0–172.16.255.255 ✓
/22: 172.16.4.0–172.16.7.255 ✓ **(longest!)**
⇒ `Interface C`

**(b) 172.16.5.200:**

Falls in /22 block (172.16.4.0–172.16.7.255), also matches /16 and /12.
⇒ `Interface C` (longest = /22)

**(c) 172.16.128.1:**

/22: No ($128 > 7$). /16: Yes ✓ **(longest)**. /12: also yes.
⇒ `Interface B`

**(d) 172.17.0.1:**

/22: No. /16: No ($172.17 \neq 172.16$). /12: 172.17 is in 172.16–172.31 ✓
⇒ `Interface A`

**(e) 10.0.0.1:**

No 172.x.x.x prefix matches. Only the default route applies.
⇒ `Interface D` (default)

## Problem 4 | Supernetting / Route Aggregation

An ISP owns four address blocks assigned to different customers:

`200.23.16.0/23` `200.23.18.0/23` `200.23.20.0/23` `200.23.22.0/23`

(a) Can these four blocks be aggregated into a single supernet? If so, what is the CIDR block?
(b) Show the binary representation of the third octets to justify your answer.
(c) If a fifth customer is assigned `200.23.24.0/23`, can all five be aggregated?

## Solution

**(b) Binary of third octets** (solving this first to justify (a)):

| Block | Third Octet Binary |
|---|---|
| 16 | 00010000 |
| 18 | 00010010 |
| 20 | 00010100 |
| 22 | 00010110 |

Each original block is /23 (9 host bits). The blocks differ only in bits 2–1 of the third octet. The top 5 bits (`00010`) are identical across all four.

**(a)** Yes! The common prefix covers $8 + 8 + 5 = 21$ bits:

$$\boxed{\texttt{200.23.16.0/21}}$$

This covers 200.23.16.0 through 200.23.23.255 ($2^{11} = 2048$ addresses).

**(c)** Adding `200.23.24.0/23`: third octet $24 = 00011000_2$.
The top 5 bits are `00011`, which differs from the others (`00010`). So we *cannot* aggregate all five into /21.

> **Could we use /20?** Matching on the top 4 bits (`0001`) gives /20, covering 200.23.16.0–200.23.31.255. But this includes addresses we don't own (200.23.24.0–200.23.31.255 minus our one /23). This wastes space and risks routing conflicts. Only safe if the ISP owns all blocks in that range.